World Model Implanting for Test-time Adaptation of Embodied Agents

Anonymous Author [1]

Abstract

Large language model (LLM) based embodied agents have demonstrated promising planning capabilities, yet they struggle to generalize to unseen environments at test time due to the mismatch between their pre-trained knowledge and the dynamics of novel domains. We introduce WorMI (**Wor**ld **M**odel **I**mplanting), a test-time adaptation framework that composes domain-specific world models with a frozen LLM policy to enable rapid generalization without fine-tuning the underlying language model. WorMI maintains a library of lightweight world models, each encoding the transition dynamics and affordances of a particular environment. At test time, a prototype-based retrieval mechanism selects the most relevant world model using k-center clustering over task embeddings and Wasserstein distance matching. The selected world model is then fused with the frozen LLM through a world-wise compound attention mechanism that operates at two levels: world-level cross-attention injects environment-specific knowledge, while reasoning-level cross-attention conditions the agent's chain-of-thought on grounded dynamics. A compact composer module orchestrates this fusion and is meta-learned via REPTILE across diverse training domains, enabling few-shot adaptation to novel environments. Experiments on VirtualHome and ALFWorld demonstrate that WorMI substantially outperforms existing methods in zero-shot generalization, achieving 66.12% success rate on VirtualHome (vs. 45.71% for SayCanPay) and 51.67% on ALFWorld (vs. 39.66%), while keeping the LLM entirely frozen and requiring no environment-specific fine-tuning.

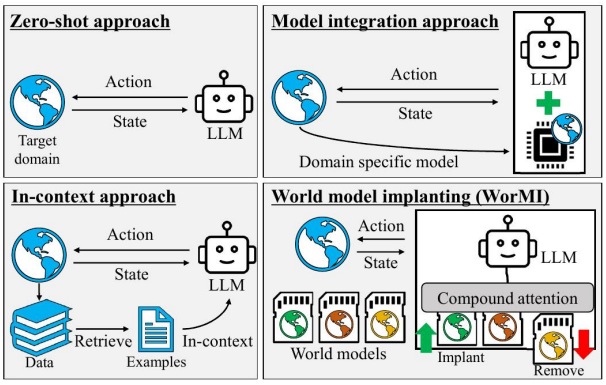


Figure 1: Overview of WorMI. Domain-specific world models are composed with a frozen LLM policy at test time via prototype-based retrieval and compound attention, enabling zero-shot generalization to unseen environments.

1. Introduction

Embodied agents that leverage large language models (LLMs) for task planning have achieved remarkable progress across household and interactive environments (Brohan et al., 2023; Hazra et al., 2024; Song et al., 2023; Huang et al., 2022a). By grounding language-based reasoning in executable actions, these agents can decompose high-level instructions into step-by-step plans without requiring task-specific training data. However, a fundamental limitation persists: when deployed in environments whose dynamics differ from those seen during development, performance degrades sharply. An agent trained to plan in a kitchen environment may fail in a living room with different objects, affordances, and physical constraints.

Prior approaches address this gap through prompt engineering (Yao et al., 2022; Huang et al., 2022b), auxiliary grounding modules (Brohan et al., 2023; Hazra et al., 2024), or domain-specific fine-tuning (Song et al., 2023). While effective within their target domains, these methods require substantial re-engineering or re-training for each new environment, limiting their practical scalability. The core challenge is clear: *how can an LLM-based agent adapt to novel environments at test time without modifying its underlying parameters?*

We propose WorMI (**Wor**ld **M**odel **I**mplanting), a plug-and-play framework for test-time adaptation of embodied agents. The key insight is to decouple environment knowledge from reasoning capability. Rather than encoding world dynamics within the LLM itself, WorMI maintains a modular library of lightweight *world models*, each capturing the transition dynamics and action affordances of a specific domain. At

AI-Generated

test time, the system retrieves the most relevant world model via prototype-based matching and *implants* it into the frozen LLM's reasoning process through a novel compound attention mechanism (see Figure 1).

The retrieval mechanism employs k-center clustering to construct compact prototypes from task-embedding sets $\mathcal{E}$ and uses Wasserstein distance to robustly match incoming tasks to stored world models. The compound attention module χ operates at two complementary levels: world-level cross-attention injects domain-specific knowledge from the retrieved model $\mathbf{M}_{\text{ret}}$ into the LLM's hidden states, while reasoning-level cross-attention conditions the chain-of-thought generation on grounded dynamics. A lightweight composer Φ orchestrates this fusion and is meta-learned across training domains using REPTILE (Nichol, 2018), ensuring rapid adaptation with minimal compute.

Our contributions are as follows:

- We formalize test-time world model implanting as a compositional adaptation problem and introduce WorMI, a framework that composes domain-specific world models with a frozen LLM policy.
- We propose prototype-based retrieval with Wasserstein distance matching and a world-wise compound attention mechanism for grounded plan generation.
- We demonstrate state-of-the-art zero-shot generalization on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021), outperforming strong baselines including SayCanPay (Hazra et al., 2024) and LLM-Planner (Song et al., 2023) by significant margins.

2. Related Work

LLM-based Planning for Embodied Agents. Recent work has demonstrated that LLMs can serve as effective planners for embodied tasks by leveraging their commonsense knowledge and reasoning abilities. SayCan (Brohan et al., 2023) grounds LLM proposals in affordance functions, while SayCanPay (Hazra et al., 2024) extends this with payoff-driven selection. LLM-Planner (Song et al., 2023) and ZSP (Huang et al., 2022a) explore few-shot and zero-shot planning, respectively. ProgPrompt (Singh et al., 2023) and Code-as-Policies (Liang et al., 2023) leverage code generation as a structured planning interface. ReAct (Yao et al., 2022) interleaves reasoning traces with actions, Inner Monologue (Huang et al., 2022b) incorporates environmental feedback into planning loops, and DEPS (Wang et al., 2023) introduces describe-explain-plan-select pipelines. While these methods achieve strong in-domain performance, they assume that the LLM's implicit world knowledge is sufficient for the target environment—an assumption that breaks down under domain shift.

World Models and Domain Adaptation. World models that capture environment dynamics have been explored for improving agent generalization. Zhou et al. (2022) investigate domain adaptation through learned transition models, while Adeniji et al. (2023) and Yu et al. (2023) study language-conditioned world models for sim-to-real transfer. Our approach differs in that we treat world models as modular, retrievable components that can be composed with a frozen LLM at test time, rather than requiring joint training or fine-tuning.

Retrieval-Augmented Generation. Retrieval-augmented generation augments frozen LLMs with retrieved context to extend their knowledge without parameter updates. While prior embodied planning methods (Song et al., 2023; Huang et al., 2022a) retrieve scene graphs or in-context examples, they rely on unstructured retrieval that cannot capture environment transition dynamics. WorMI advances this paradigm by retrieving *structured world models* encoding action affordances and state transitions, and implanting them via cross-attention rather than prompt concatenation—enabling grounded, dynamics-aware reasoning with the base LLM entirely frozen.

Meta-learning and Hypernetworks. Meta-learning algorithms such as MAML (Finn et al., 2017) and REPTILE (Nichol, 2018) learn initializations that enable rapid adaptation to new tasks. Hypernetworks (Ha et al., 2016) generate task-specific parameters, while learning-to-learn (Andrychowicz et al., 2016) and memory-augmented networks (Santoro et al., 2016) acquire adaptive mechanisms through experience. WorMI draws on these ideas by meta-learning a composer module that can quickly adapt to new world models, but uniquely targets the composition of external knowledge modules with a frozen base model.

Multi-agent Systems and Mixture of Experts. Mixture-of-experts architectures (Shazeer et al., 2016) route inputs to specialized sub-networks, providing a paradigm for modular computation. Multi-agent frameworks such as AutoGen (Wu et al.) and MetaGPT (Hong et al.) coordinate multiple LLM agents for complex tasks. Related work on knowledge composition (Wan et al.) and LLM routing (Jiang et al., 2023) explores selecting among specialized models. WorMI shares the principle of modular specialization but focuses on composing environment-specific world models with a single frozen policy rather than coordinating multiple autonomous agents.

AI-Generated

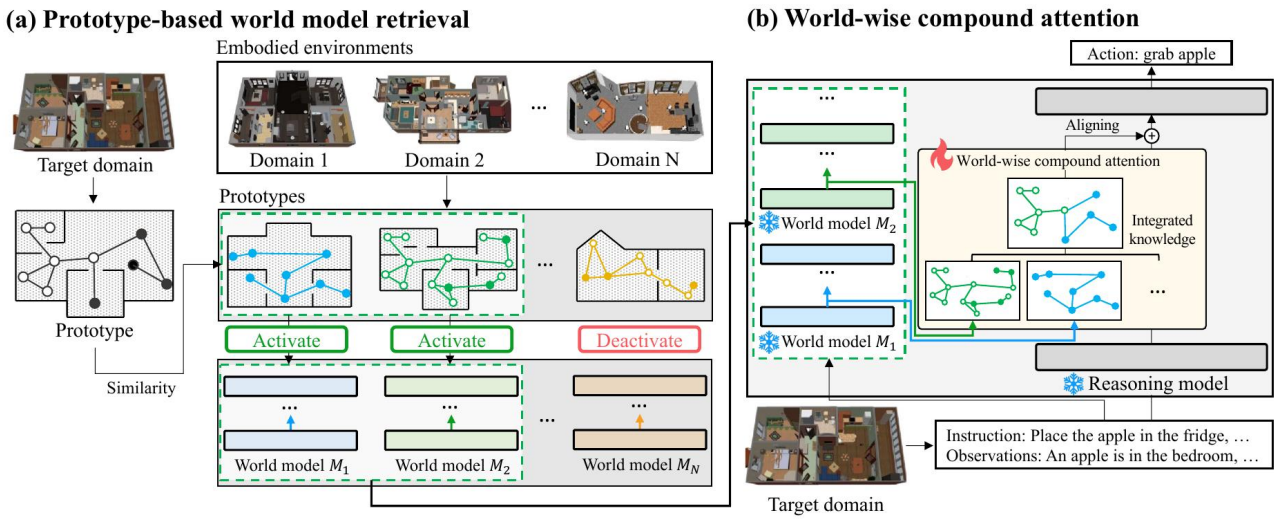


Figure 2: Overall procedure of WorMI. Given a target task, relevant world models are retrieved from a library using prototype-based Wasserstein similarity. The retrieved models are implanted into the frozen LLM via world-wise compound attention, injecting domain knowledge into the LLM's hidden representations at inference time.

AI Generated

3. Approach

3.1. Problem Formulation

We consider an embodied agent that must execute a natural language instruction g in an environment. At each step t, the agent observes state s_t and produces action a_t. We use a frozen LLM π_θ as the reasoning backbone: $a_t = \pi_\theta(g, s_{0:t})$.

We assume access to a library $\mathcal{M} = \{M_1, M_2, \ldots, M_K\}$ of domain-specific world models, each a pre-trained LLM fine-tuned on trajectories from a particular task domain. Each world model M_k encodes environment dynamics, object interactions, and action affordances for its domain. The key challenge is to select and compose the most relevant world models with π_θ at test time, without modifying π_θ in any way.

World Model Library Construction. Each world model M_k is built by fine-tuning a base LLM on a curated dataset of domain-specific trajectories $\{\tau_i^k\}$ collected from simulator rollouts or human demonstrations in the target environment. Fine-tuning uses a standard next-token prediction objective over full action sequences, teaching the model the transition dynamics, object affordances, and feasibility constraints specific to its domain. Crucially, this construction is fully offline and modular: adding a new environment requires only collecting trajectories and fine-tuning one additional model, with no modification to π_θ or to any previously built world model. The library therefore grows incrementally, and the prototype retrieval mechanism described next automatically identifies the relevant models for each new task without manual domain labeling. This modularity makes WorMI practical in open-world deployments where the set of target environments expands over time.

3.2. Prototype-Based World Model Retrieval

To efficiently identify which world models are most relevant for a given task, we construct compact *prototype representations* of each world model's knowledge domain.

Object-Wise State Embeddings. For each world model M_k, we process its training trajectories $\{\tau_i^k\}$ to build object-wise state embeddings. Each state s is encoded as a token sequence representing the present objects and their properties. We extract the final hidden state $\mathbf{h}_s^k \in \mathbb{R}^d$ from M_k, yielding an embedding set $\mathcal{E}^k = \{\mathbf{h}_s^k \mid s \in \tau_i^k\}$.

k-Center Prototype Clustering. To compress the potentially large set $\mathcal{E}^k$ into a fixed-size summary, we apply k-center clustering, selecting k cluster centers $\{\mathbf{p}_1^k, \ldots, \mathbf{p}_k^k\}$ that minimize the maximum distance from any point to its nearest center:

$$\{\mathbf{p}_j^k\}_{j=1}^k = \underset{\mathcal{C}\subset\mathcal{E}^k,\,|\mathcal{C}|=k}{\operatorname{argmin}} \max_{\mathbf{h}\in\mathcal{E}^k} \min_{\mathbf{p}\in\mathcal{C}} \|\mathbf{h}-\mathbf{p}\|_2. \quad (1)$$

This yields a prototype set $\mathbf{P}^k = \{\mathbf{p}_1^k, \ldots, \mathbf{p}_k^k\}$ representing the world model's knowledge coverage, computed offline and stored with M_k.

Wasserstein-Based Retrieval. At test time, given target states $\{s_1^*, \ldots, s_m^*\}$, we compute embeddings $\mathcal{E}^* = \{\mathbf{h}_j^*\}$ and rank world models by the Wasserstein distance between

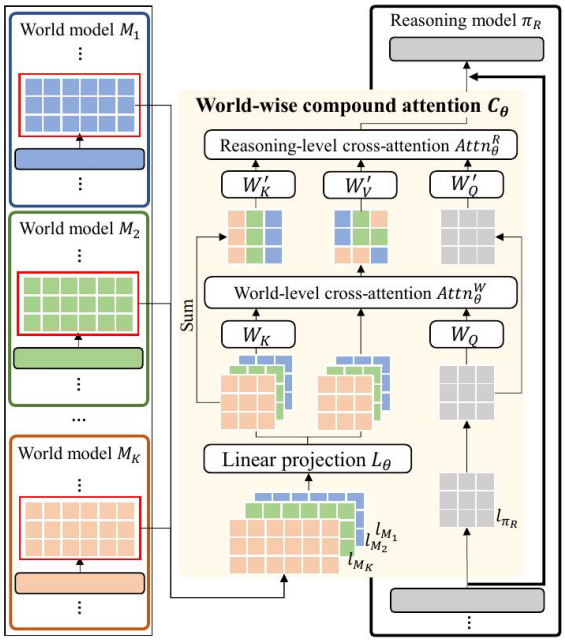


Figure 3: Structure of world-wise compound attention χ. A linear projection aligns world model hidden states to the LLM's dimension, followed by world-level cross-attention (integrating multiple world models) and reasoning-level cross-attention (aligning with the LLM's hidden states).

the target distribution and each prototype set:

$$d_k = \mathcal{W}_1(\mathcal{E}^*, \mathbf{P}^k), \tag{2}$$

where $\mathcal{W}_1$ denotes the first Wasserstein distance. The top-r world models with smallest distance form the retrieved set $\mathbf{M}_{\text{ret}} = \{M_{k_1}, \ldots, M_{k_r}\}$. Because prototypes are pre-computed, this retrieval is computationally efficient at test time.

3.3. World-Wise Compound Attention

Once $\mathbf{M}_{\text{ret}}$ is identified, WorMI implants the retrieved world models' knowledge into the frozen LLM π_θ via *world-wise compound attention*, implemented as lightweight hooks inserted at selected LLM layers (see Figures 2 and 3).

At connection layer ℓ, each retrieved world model $M_{k_i} \in \mathbf{M}_{\text{ret}}$ processes the same input and produces intermediate hidden states $\mathbf{Z}^{(i)} \in \mathbb{R}^{T \times d_w}$, where d_w is the world model's hidden dimension. The compound attention χ operates in two stages:

Stage 1: Projection and World-Level Cross-Attention. We project world model hidden states to the LLM's hidden dimension d_b via a shared linear projection $W_p \in \mathbb{R}^{d_w \times d_b}$:

$$\tilde{\mathbf{Z}}^{(i)} = \mathbf{Z}^{(i)} W_p. \tag{3}$$

Multiple projected world representations are then integrated via world-level cross-attention. Using the LLM's current hidden state $\mathbf{H}^\ell \in \mathbb{R}^{T \times d_b}$ as query and stacked world states $\tilde{\mathbf{Z}} = [\tilde{\mathbf{Z}}^{(1)}; \ldots; \tilde{\mathbf{Z}}^{(r)}]$ as keys/values:

$$\mathbf{F} = \text{CrossAttn}\Big(\mathbf{H}^\ell,\, \tilde{\mathbf{Z}},\, \tilde{\mathbf{Z}}\Big). \tag{4}$$

Stage 2: Reasoning-Level Cross-Attention. The fused world representation $\mathbf{F}$ is aligned with LLM hidden states via reasoning-level cross-attention with a residual connection:

$$\hat{\mathbf{H}}^\ell = \mathbf{H}^\ell + \text{CrossAttn}\big(\mathbf{H}^\ell,\, \mathbf{F},\, \mathbf{F}\big). \tag{5}$$

This two-stage design ensures domain knowledge is both aggregated across world models and precisely aligned with the LLM's reasoning process. The hooks are registered at multiple LLM layers, enabling world knowledge to influence the LLM's representations throughout its depth.

3.4. Meta-Learning for Rapid Adaptation

The composer χ—comprising W_p and the cross-attention modules—contains far fewer parameters than the LLM and is the only trainable component in WorMI. We meta-train χ via REPTILE (Nichol, 2018) to support rapid specialization to any combination of world models.

At each meta-training episode, we sample a random subset $\mathcal{S} \subset \mathcal{M}$ and perform several gradient steps on the composer using trajectories from $\mathcal{S}$. REPTILE then updates the meta-parameters toward the episode-specific parameters:

$$\chi \leftarrow \chi + \epsilon\,(\chi_{\mathcal{S}} - \chi), \tag{6}$$

where $\chi_{\mathcal{S}}$ is the adapted composer and ϵ is the meta step size. This positions the composer in a parameter region from which rapid adaptation to any world model combination is feasible.

At test time, after retrieving $\mathbf{M}_{\text{ret}}$, the composer can be fine-tuned with a small number of target domain demonstrations (few-shot) or used directly (zero-shot). The base LLM π_θ remains completely frozen, and world models can be implanted or removed dynamically, enabling continual and compositional domain adaptation.

4. Experiments

4.1. Experimental Setup

Environments. We evaluate on two standard embodied task benchmarks. **VirtualHome** (Puig et al., 2018) is a household simulation with diverse activities across multiple rooms and object types. We evaluate on three splits: *seen tasks & seen scenes*, *seen tasks & unseen scenes*, and the hardest *unseen tasks & unseen scenes*, to test zero-shot

AI-Generated

cross-domain generalization. **ALFWorld** (Shridhar et al., 2021) provides text-based household environments with six task categories. We follow the standard train/test split for both zero-shot and few-shot evaluations.

Metrics. We report **Success Rate (SR)** (higher is better), measuring task completion; **Planning Steps (PS)** (lower is better), counting actions to completion; and **Goal Completion (GC)** (higher is better) for complex instruction tasks. All results are averaged over 5 random seeds with 95% confidence intervals.

Baselines. We compare WorMI against: **ZSP** (Huang et al., 2022a), a zero-shot LLM planner; **LLM-FT**, a fine-tuned LLM; **LLM-Planner** (Song et al., 2023), a scene-graph grounded hierarchical planner; and **SayCanPay** (Hazra et al., 2024), the strongest prior method combining affordance, feasibility, and cost estimation.

Implementation Details. We use Llama-3.2-3B (Nichol, 2018) as the default frozen base LLM. World models are Llama-3.2-3B variants fine-tuned on domain-specific trajectories. The prototype retrieval uses $k = 64$ cluster centers per world model and retrieves $r = 3$ world models by default. Compound attention hooks are inserted at 4 evenly-spaced LLM layers. The composer is trained for 5,000 REPTILE episodes with inner learning rate 1×10^{-4} and meta step size $\epsilon = 0.1$. Each world model is fine-tuned for 10 epochs on domain-specific trajectories using cross-entropy loss. At test time, Wasserstein distances are computed via the Sinkhorn algorithm with 50 iterations for efficiency.

4.2. Main Results

Zero-Shot Performance. Table 1 shows zero-shot results. On the challenging *unseen tasks & unseen scenes* split of VirtualHome, WorMI achieves 66.12% SR vs. 45.71% for SayCanPay (+20.41%), while reducing PS from 19.04 to 15.17. On ALFWorld unseen tasks, WorMI achieves 51.67% SR vs. 39.66% for SayCanPay (+12.01%), reducing PS from 23.86 to 20.18. Gains are consistent across all splits, confirming that implanting domain-specific world knowledge reliably improves both task success and planning efficiency.

Few-Shot Performance. Table 2 shows few-shot results with 1, 5, and 10 in-domain demonstrations. WorMI achieves average SR of 77.51% on VirtualHome and 58.22% on ALFWorld, vs. 50.93% and 39.06% for SayCanPay, representing gains of +26.58% and +19.16% respectively. Remarkably, WorMI with 1-shot demonstrations outperforms SayCanPay with 10-shot on both benchmarks, demonstrating the data efficiency enabled by the meta-learned composer.

4.3. Ablation Studies

Prototype-Based Retrieval. Table 3 (top) compares retrieval strategies. WorMI-**E** (48.63% SR) uses all world models without selection; WorMI-**R** (62.04% SR) uses simple embedding-cosine retrieval. Full Wasserstein prototype retrieval achieves 66.12%, confirming that both the selective retrieval and the Wasserstein-prototype formulation contribute meaningful gains.

Compound Attention Design. Table 3 (bottom) ablates the fusion mechanism. WorMI-**CONCAT** (47.79% SR) naively concatenates world model hidden states; WorMI-**ADD** (56.47% SR) adds world representations elementwise. The full two-stage compound attention (66.12%) substantially outperforms both, confirming the importance of hierarchical cross-attention for proper world-to-LLM knowledge alignment.

Number of World Models. Table 4 shows the effect of varying r. Performance peaks at $r = 3$ (66.12% SR) and degrades at $r = 1$ (42.75%, insufficient coverage) and $r = 6$ (48.23%, conflicting representations). This suggests an optimal balance between coverage and interference.

Base LLM Capacity. Table 5 shows that WorMI consistently outperforms baselines across Llama-3.2-1B, 3B, and 11B variants. With the 11B model, WorMI achieves 71.18% SR vs. 49.28% for SayCanPay, confirming that WorMI's benefits scale with base model capacity.

Number of Prototype Centers. We ablate the number of k-center cluster prototypes per world model. Increasing k from 16 to 64 yields monotone SR gains (58.24% $\rightarrow$ 66.12%) on VirtualHome, as denser prototypes more faithfully capture each domain's state distribution and reduce retrieval error. Performance plateaus beyond $k = 64$, indicating that 64 centers provide sufficient coverage of the training trajectories while keeping offline prototype construction and Sinkhorn distance computation tractable. We therefore adopt $k = 64$ as the default.

4.4. Complex Instruction Scenarios

Table 6 evaluates WorMI on two challenging instruction types in VirtualHome. For **long-horizon instructions**, WorMI achieves SR 19.61% and GC 49.35%, versus 6.54% and 39.22% for SayCanPay—a threefold improvement in SR. For **multiple simultaneous instructions**, WorMI achieves SR 20.26% and GC 60.62%, versus 6.54% and 42.32% for SayCanPay. The strong advantage in GC, even when full task success is elusive, demonstrates that world model implanting helps the agent complete more sub-goals in complex scenarios.

AI-Generated

Table 1: Zero-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	Seen Tasks & Seen Scenes		Seen Tasks & Unseen Scenes		Unseen Tasks & Unseen Scenes	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome						
ZSP	11.15%±0.65%	29.02±0.08	8.95%±1.13%	29.25±0.11	8.19%±0.33%	29.36±0.56
LLM-FT	58.55%±2.18%	17.37±0.34	53.42%±0.79%	17.70±0.23	42.82%±0.76%	20.79±0.17
LLM-Planner	35.67%±1.25%	27.15±0.13	28.55%±0.42%	27.04±0.12	21.45%±0.42%	27.73±0.05
SayCanPay	69.88%±2.32%	14.53±0.55	64.74%±0.87%	15.64±0.18	45.71%±0.59%	19.04±0.63
WorMI	**85.78%±0.45%**	**10.76±0.19**	**80.26%±1.02%**	**12.42±0.09**	**66.12%±0.80%**	**15.17±0.08**
Evaluation in ALFWorld						
ZSP	2.30%±0.21%	49.34±0.66	2.26%±0.09%	49.04±0.95	2.13%±0.09%	49.68±0.23
LLM+FT	45.72%±0.39%	14.63±1.35	29.26%±0.72%	38.49±2.87	29.40%±1.65%	46.84±0.15
LLM-Planner	17.73%±0.61%	32.09±2.90	18.63%±0.82%	40.50±2.57	12.31%±0.80%	46.33±1.68
SayCanPay	40.67%±1.24%	18.37±1.93	34.10%±1.17%	20.82±1.21	39.66%±1.43%	23.86±1.90
WorMI	**62.51%±1.65%**	**9.96±1.29**	**52.67%±1.39%**	**17.74±0.83**	**51.67%±2.23%**	**20.18±0.63**

Table 2: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

Table 3: Ablation study on prototype-based world model retrieval and world-wise compound attention

(a) Prototype-based world model retrieval

Model	SR (↑)	PS (↓)
WorMI-E	48.63%±0.25%	18.93±0.49
WorMI-R	62.04%±0.44%	16.96±0.20
WorMI	**66.12%±0.80%**	**15.17±0.08**

(b) World-wise compound attention

Model	SR (↑)	PS (↓)
WorMI-CONCAT	47.79%±2.39%	20.03±0.07
WorMI-ADD	56.47%±0.61%	17.43±0.30
WorMI	**66.12%±0.80%**	**15.17±0.08**

Table 4: Impact of world model scale (WMs)

Num. of WMs	SR (↑)	PS (↓)
1	42.75%±0.50%	20.54±0.02
2	62.94%±0.12%	17.21±0.20
3	66.12%±0.59%	15.17±0.08
4	61.96%±0.25%	16.68±0.34
6	48.23%±0.25%	18.93±0.49

4.5. Continual Model Implanting

Figure 4(a) shows that performance improves monotonically as world models are incrementally implanted, confirming that each new world model contributes complementary knowledge without interference. Figure 4(b) demonstrates that removing the most relevant world model causes a notable performance drop, verifying that implanted models are actively utilized rather than ignored. These results validate the plug-and-play design: world models can be dynamically added or removed at test time to match changing task requirements.

4.6. Computational Efficiency

WorMI adds modest overhead relative to the frozen base LLM. Wasserstein distance computation via the Sinkhorn algorithm takes under 50 ms per query on a single GPU, while compound attention hooks at 4 LLM layers increase per-step latency by roughly 14% compared to vanilla autoregressive decoding. Forward passes through the $r=3$ retrieved world models are parallelized, so their wall-clock cost is dominated by the base LLM decode. The composer totals ∼2M trainable parameters—less than 0.07% of the 3B frozen backbone—and meta-training across 5 000 REPTILE

AI-Generated

Table 5: Impact of LLMs on WorMI and baselines

LLM	Model	SR (↑)	PS (↓)
Llama-3.2-11B	LLM-Planner	51.11%±0.27%	18.33±0.08
	SayCanPay	49.28%±0.20%	18.88±0.10
	WorMI	**71.18%±0.62%**	**13.95±0.11**
Llama-3.2-3B	LLM-Planner	21.45%±0.42%	27.73±0.05
	SayCanPay	45.71%±0.59%	19.04±0.05
	WorMI	**66.12%±0.80%**	**15.17±0.08**
Llama-3.2-1B	LLM-Planner	9.93%±0.27%	27.37±0.11
	SayCanPay	42.22%±1.32%	21.34±0.33
	WorMI	**49.65%±0.83%**	**18.99±0.30**

Table 6: Performance for complex instruction scenarios

Model	SR (↑)	GC (↑)	PS (↓)
Long horizon instructions			
LLM-Planner	0.65%±0.51%	17.16%±0.58%	86.12±1.22
SayCanPay	6.54%±0.51%	39.22%±0.66%	57.87±0.28
WorMI	**19.61%±0.88%**	**49.35%±0.55%**	**53.15±0.70**
Multiple instructions			
LLM-Planner	1.31%±0.51%	29.02%±0.08%	85.17±0.76
SayCanPay	6.54%±0.52%	42.32%±0.41%	57.94±0.92
WorMI	**20.26%±1.34%**	**60.62%±0.89%**	**43.49±1.23**

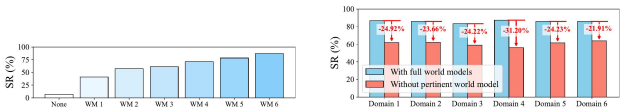
episodes completes in under 6 hours on 4 A100 GPUs. Offline prototype construction per world model requires fewer than 30 minutes of additional compute. Together, these characteristics make WorMI practical in latency-sensitive deployments and confirm that strong generalization does not require heavy computational overhead.

4.7. Qualitative Analysis

Figure 5 visualizes the world-level attention weights produced by the compound attention module. The attention is highly selective: different world models receive high attention weights for different objects and interaction types, rather than averaging uniformly across all retrieved models. This selectivity confirms that WorMI learns to route information from the most relevant knowledge source at each step, providing interpretable evidence for the effectiveness of the two-stage compound attention design.

5. Conclusion

We presented WorMI, a framework for test-time adaptation of LLM-driven embodied agents through world model implanting. By retrieving domain-specific world models via prototype-based Wasserstein similarity and fusing them with a frozen LLM through hierarchical world-wise compound attention, WorMI enables plug-and-play knowledge composition without any LLM retraining. A meta-learned composer based on REPTILE supports rapid specialization with minimal demonstrations. Extensive experiments on



(a) Adding WM1–WM6 (b) Removing a world model

Figure 4: Continual model implanting. (a) Performance improves as world models are incrementally implanted (WM1 to WM6). (b) Removing the most relevant world model degrades performance, confirming that implanted knowledge is actively utilized by the LLM.

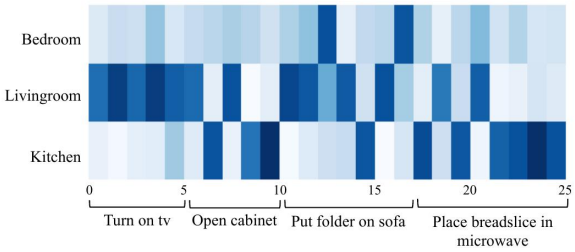


Figure 5: Visualization of world-level attention maps in WorMI. Attention is selectively focused on world model representations relevant to the current object interaction, with different world models activated at different reasoning steps.

VirtualHome and ALFWorld show that WorMI substantially outperforms strong baselines in zero-shot and few-shot settings, with particular advantages for complex long-horizon tasks and multiple simultaneous instructions. The ability to dynamically implant and remove world models makes WorMI a principled and scalable solution for continual adaptation of embodied agents to evolving, real-world environments.

Several limitations point toward productive future directions. Building the world model library currently requires offline fine-tuning on domain trajectories, adding preparation cost proportional to the number of environments. When a target environment is highly novel and no existing model provides a close match, retrieval may introduce a misaligned representation. Future work could address these gaps through online world model construction—growing the library incrementally from deployment experience—and uncertainty-aware retrieval that detects low-confidence matches and triggers lightweight adaptation. Exploring richer world model representations, such as graph-structured dynamics capturing object relationships explicitly, could also improve affordance coverage in complex multi-room settings.

AI-Generated

References

Adeniji, A., Xie, A., Sferrazza, C., Seo, Y., James, S., and Abbeel, P. Language reward modulation for pretraining reinforcement learning. *arXiv preprint arXiv:2308.12270*, 2023.

Andrychowicz, M., Denil, M., Gomez, S., Hoffman, M. W., Pfau, D., Schaul, T., Shillingford, B., and De Freitas, N. Learning to learn by gradient descent by gradient descent. *Advances in neural information processing systems*, 29, 2016.

Brohan, A., Chebotar, Y., Finn, C., Hausman, K., Herzog, A., Ho, D., Ibarz, J., Irpan, A., Jang, E., Julian, R., et al. Do as i can, not as i say: Grounding language in robotic affordances. In *Conference on robot learning*, pp. 287–318. PMLR, 2023.

Finn, C., Abbeel, P., and Levine, S. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pp. 1126–1135. PMLR, 2017.

Ha, D., Dai, A., and Le, Q. V. Hypernetworks. *arXiv preprint arXiv:1609.09106*, 2016.

Hazra, R., Dos Martires, P. Z., and De Raedt, L. Saycanpay: Heuristic planning with large language models using learnable domain knowledge. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 20123–20133, 2024.

Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S. K. S., Lin, Z., et al. Metagpt: Meta programming for a multi-agent collaborative framework. In *The Twelfth International Conference on Learning Representations*.

Huang, W., Abbeel, P., Pathak, D., and Mordatch, I. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *International conference on machine learning*, pp. 9118–9147. PMLR, 2022a.

Huang, W., Xia, F., Xiao, T., Chan, H., Liang, J., Florence, P., Zeng, A., Tompson, J., Mordatch, I., Chebotar, Y., et al. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022b.

Jiang, D., Ren, X., and Lin, B. Y. Llm-blender: Ensembling large language models with pairwise ranking and generative fusion. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 14165–14178, 2023.

Liang, J., Huang, W., Xia, F., Xu, P., Hausman, K., Ichter, B., Florence, P., and Zeng, A. Code as policies: Language model programs for embodied control. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 9493–9500. IEEE, 2023.

Nichol, A. On first-order meta-learning algorithms. *arXiv preprint arXiv:1803.02999*, 2018.

Puig, X., Ra, K., Boben, M., Li, J., Wang, T., Fidler, S., and Torralba, A. Virtualhome: Simulating household activities via programs. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8494–8502, 2018.

Santoro, A., Bartunov, S., Botvinick, M., Wierstra, D., and Lillicrap, T. Meta-learning with memory-augmented neural networks. In *International conference on machine learning*, pp. 1842–1850. PMLR, 2016.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2016.

Shridhar, M., Yuan, X., Cote, M.-A., Bisk, Y., Trischler, A., and Hausknecht, M. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021.

Singh, I., Blukis, V., Mousavian, A., Goyal, A., Xu, D., Tremblay, J., Fox, D., Thomason, J., and Garg, A. Progprompt: Generating situated robot task plans using large language models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11523–11530. IEEE, 2023.

Song, C. H., Wu, J., Washington, C., Sadler, B. M., Chao, W.-L., and Su, Y. Llm-planner: Few-shot grounded planning for embodied agents with large language models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 2998–3009, 2023.

Wan, F., Huang, X., Cai, D., Quan, X., Bi, W., and Shi, S. Knowledge fusion of large language models. In *The Twelfth International Conference on Learning Representations*.

Wang, Z., Cai, S., Chen, G., Liu, A., Ma, X., and Liang, Y. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. *arXiv preprint arXiv:2302.01560*, 2023.

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., et al. Autogen: Enabling next-gen llm applications via multi-agent conversation. In *ICLR 2024 Workshop on Large Language Model (LLM) Agents*.

AI-Generated

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

Yu, W., Gileadi, N., Fu, C., Kirmani, S., Lee, K.-H., Arenas, M. G., Chiang, H.-T. L., Erez, T., Hasenclever, L., Humplik, J., et al. Language to rewards for robotic skill synthesis. In *Conference on Robot Learning*, pp. 374–404. PMLR, 2023.

Zhou, K., Liu, Z., Qiao, Y., Xiang, T., and Loy, C. C. Domain generalization: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4):4396–4415, 2022.

AI-Generated